6. Lineární lomená funkce a mocninné funkce (MO 07)

nepřímá úměrnost jako funkce

lineární lomená funkce, definiční obor, obor hodnot, nalezení asymptot

průsečíky s osami souřadnic

mocninná funkce (kladný, záporný, sudý, lichý exponent, racionální exponent)

Teorie + tabulky:

nepřímá úměrnost $y = \frac{k}{x}$

- grafem rovnoosá hyperbola

$D = R - \{0\}$ asymptoty $x = 0$

$H = R - \{0\}$ $y = 0$

lin. lom. funkce

$f(x) = \frac{ax+b}{cx+d}$ $c \neq 0$

$cx + d \neq 0$

$x \neq -\frac{d}{c} \rightarrow D = R - \{-\frac{d}{c}\}$

$a \parallel y \rightarrow x = -\frac{d}{c}$

$a \parallel x \rightarrow y = \frac{a}{c}$

Příklady, které mi nešly:

Co mi nejde + dotazy:

1. Sestrojte grafy funkcí, určete D_f, H_f a průsečíky s osami:

a) $y = \frac{2x-6}{3x+5}$

$3x \neq -5$

$x \neq -\frac{5}{3}$

$D = R - \{-\frac{5}{3}\}$

$y = \frac{2}{3}$

$H = R - \{\frac{2}{3}\}$

$P_y [0; \frac{-6}{5}]$

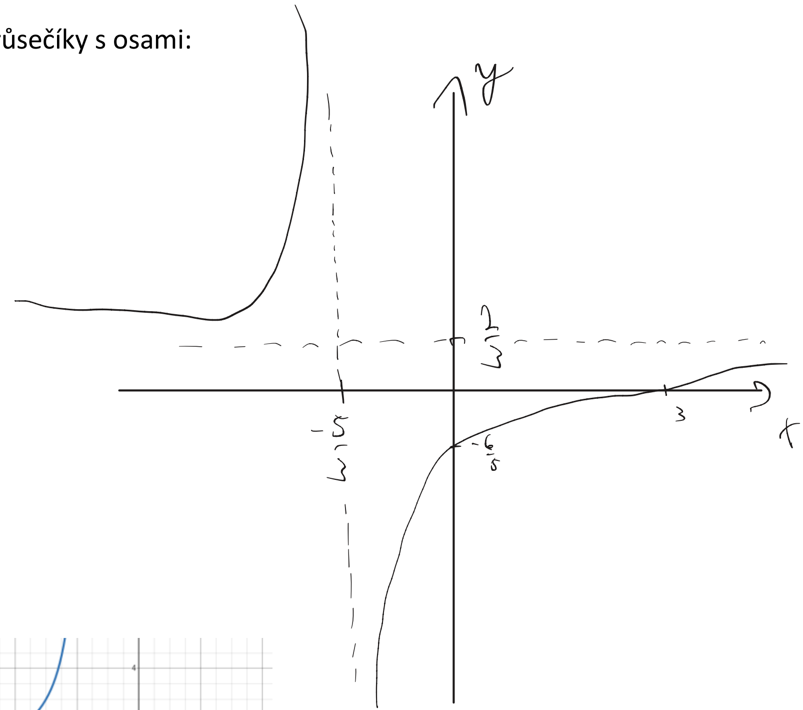
$y = \frac{-6}{5}$

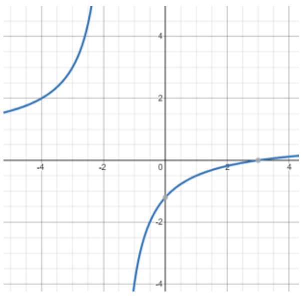
$P_x [3; 0]$

$0 = \frac{2x-6}{3x+5}$

$6 = 2x$

$x = 3$





[a)] $D_f = \boldsymbol{R} - \left\{-\frac{5}{3}\right\}$, $H_f = \boldsymbol{R} - \left\{\frac{2}{3}\right\}$, $P_x[3;0]$, $P_y\left[0; -\frac{6}{5}\right]$]

b) $y = \frac{3x}{x-2}$

$x \neq 2$

$D = R - \{2\}$

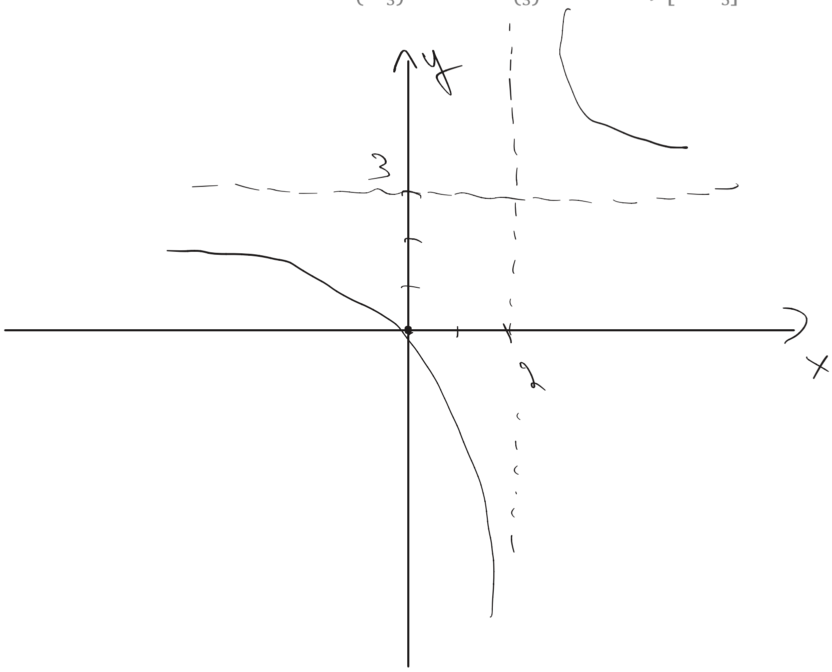
$H = R - \{3\}$

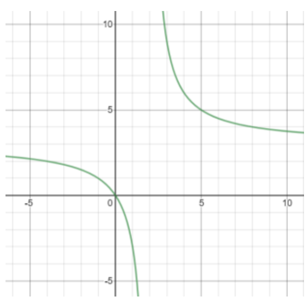
$P_x [0; 0]$

$P_y [0; 0]$

$0 = \frac{3x}{x-2}$

$x = 0$





[b)] $D_f = \boldsymbol{R} - \{2\}$, $H_f = \boldsymbol{R} - \{3\}$, $P_x = P_y = [0;0]$]

c) $y = \left|\dfrac{3x}{2-x}\right|$

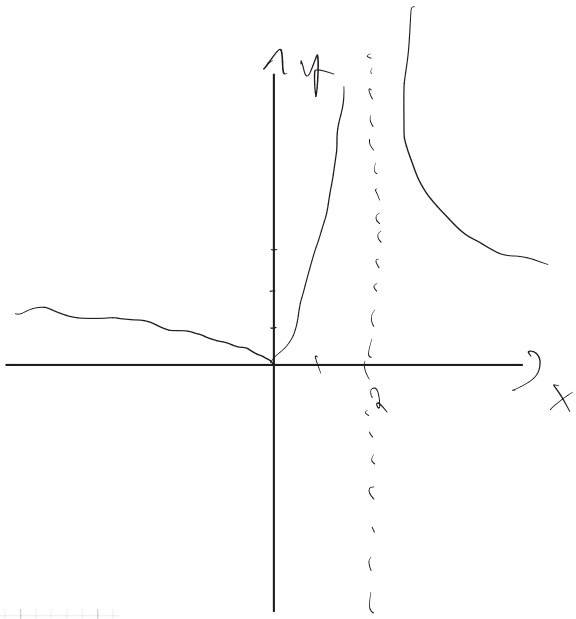
$x \neq 2$

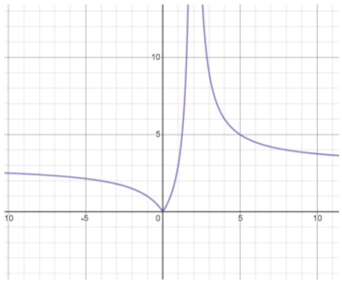
$D = \mathbb{R} - \{2\}$

$H = \langle 0; \infty)$

$P_x [0;0]$

$P_y [0;0]$





[c) $D_f = \boldsymbol{R} - \{2\}$, $H_f = \langle 0;\infty)$, $P_x = P_y = [0;0]$]

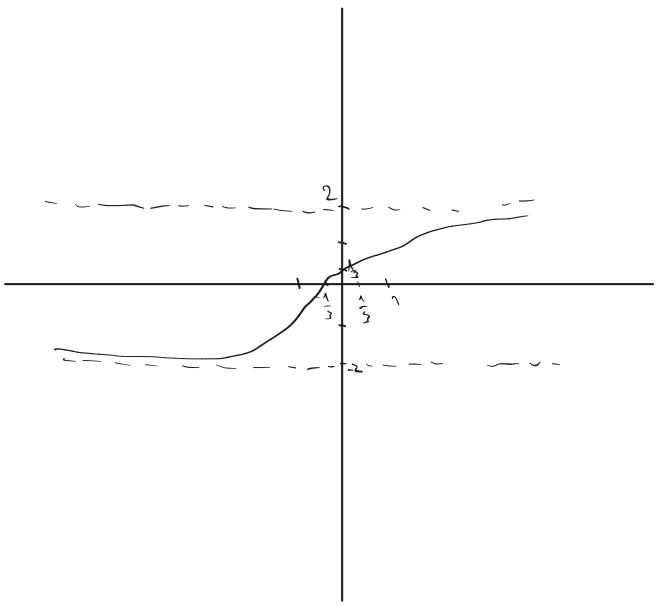
d) $y = \dfrac{2x+1}{|x|+3}$

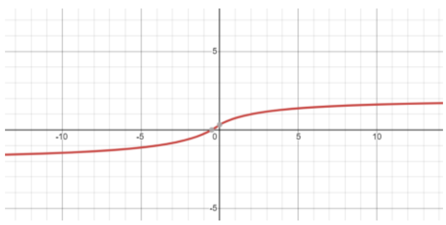
$D = \mathbb{R}$

$H = (-2;2)$

$P_x [-\frac{1}{2};0]$

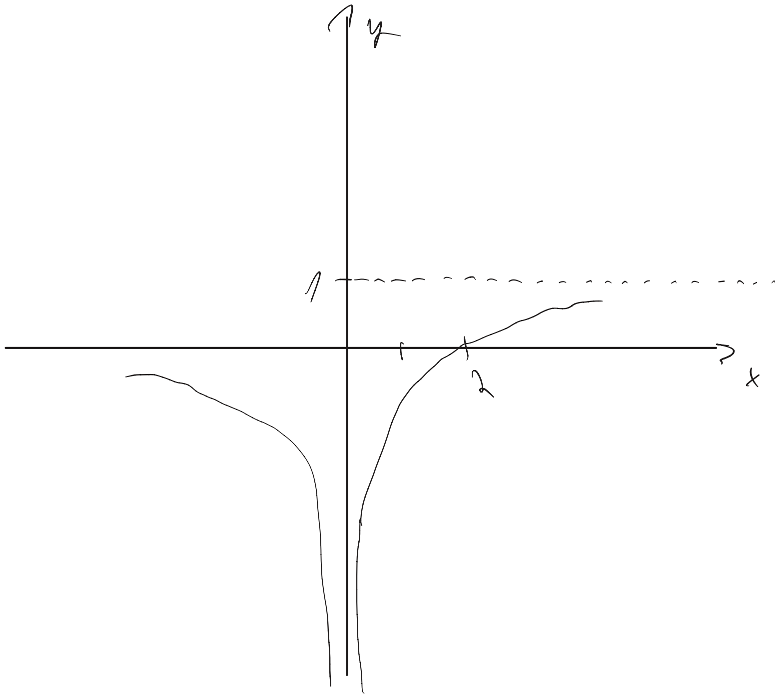
$P_y [0;\frac{1}{3}]$





[d) $D_f = \boldsymbol{R}$, $H_f = (-2;2)$, $P_x\left[-\frac{1}{2};0\right]$, $P_y\left[0;\frac{1}{3}\right]$]

e) $y = \dfrac{x-2}{|x|}$



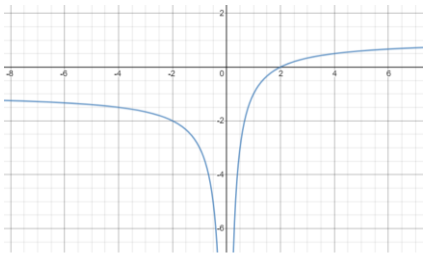
$D = R - \{0\}$

$H = (-\infty;1)$

$P_x[2;0]$

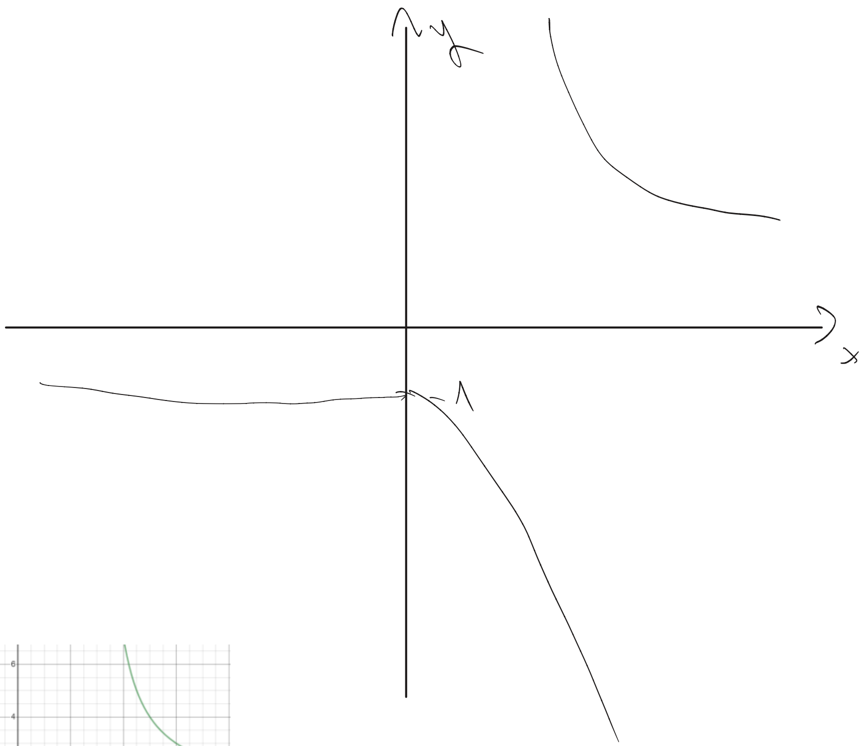
P_y neexistuje

$x \neq 0$

[e] 

$D_f = \boldsymbol{R} - \{0\}$, $H_f = (-\infty; 1)$, $P_x[2; 0]$, P_y neex.]

f) $y = \dfrac{x+3}{|x|-3}$



$D = R - \{\pm 3\}$

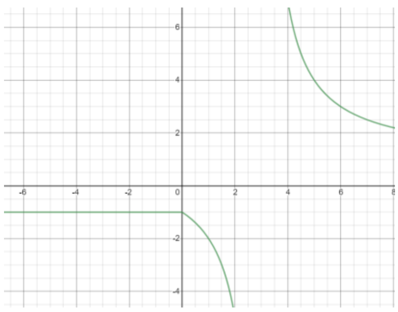
$H = (-\infty;1\rangle \cup (1;\infty)$

$x+3=0$

$x=-3$

P_x neexistuje

$P_y[0;-1]$

[f] 

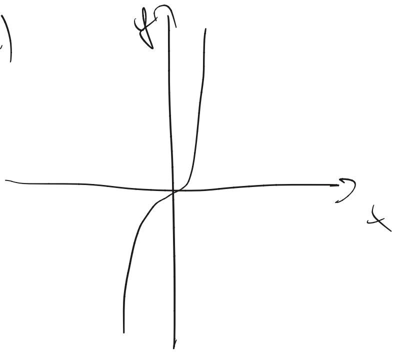
$D_f = \boldsymbol{R} - \{\pm 3\}$, $H_f = (-\infty;-1\rangle \cup (1;\infty)$, P_x neex., $P_y[0;-1]$]

2. Načrtněte grafy funkcí:

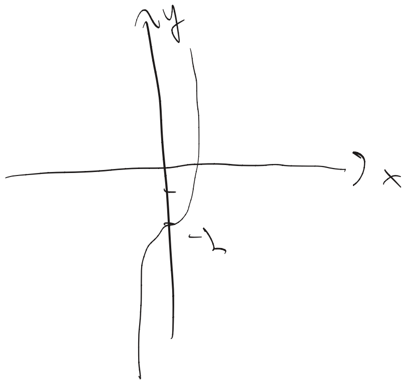
↓2 →1 ↑2

a) $y = x^3$ b) $y = x^3 - 2$ c) $y = (x-1)^3 + 2$ d) $y = 2x^3$

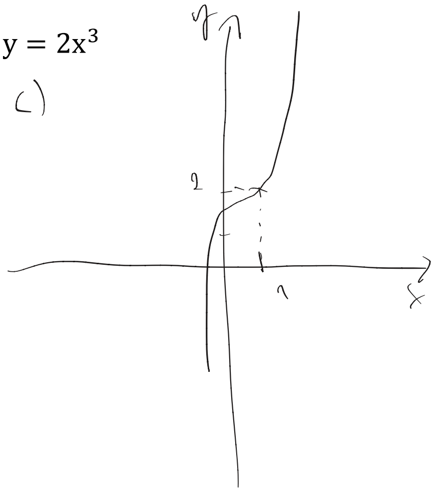
a)



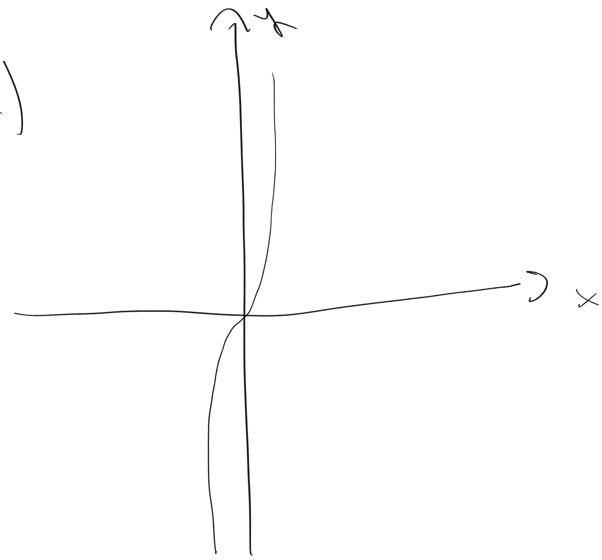
b)

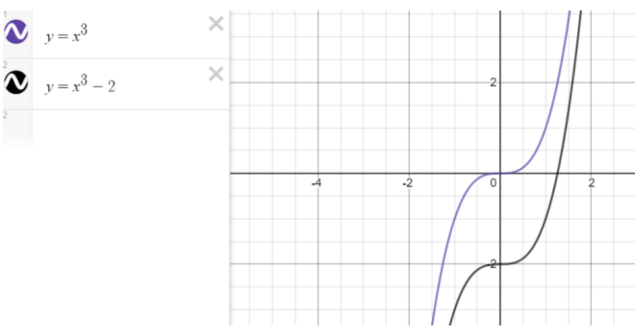


c)

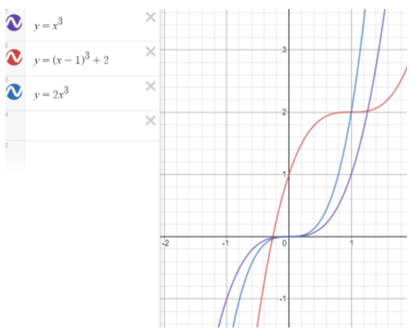


d)





[a) +b)

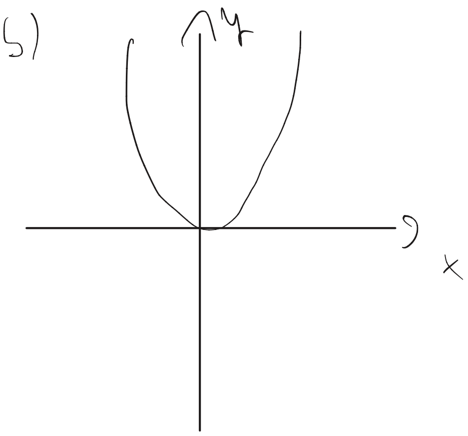


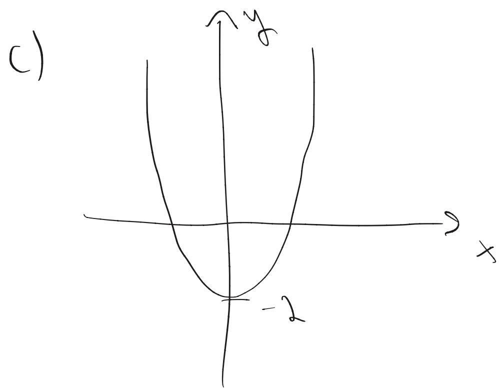
c)+d)

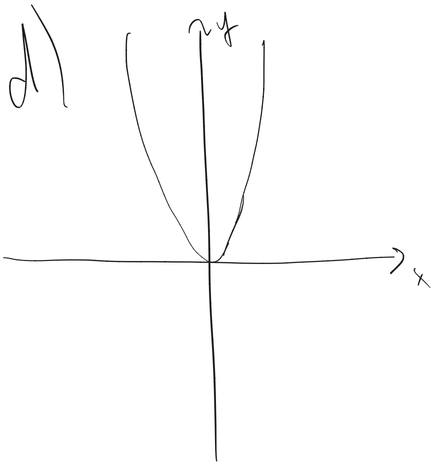
]

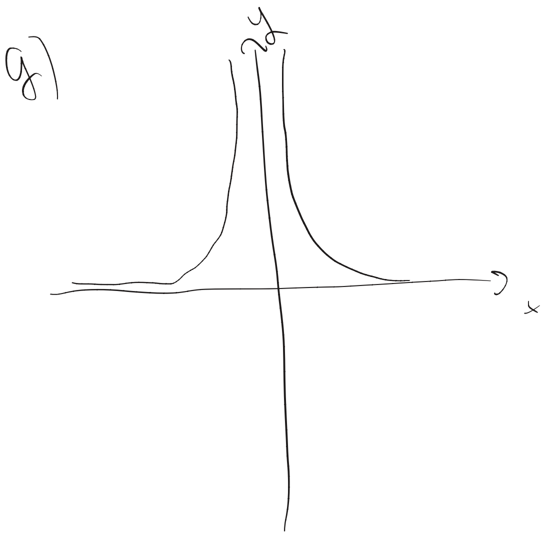
3. Vyberte sudé funkce a načrtněte jejich graf:

a) $y = x^{-3}$
b) $y = x^2 - 2$ $\downarrow 2$
c) $y = x^4$
d) $y = 3x^4$
e) $y = x^5$
f) $y = (x-1)^4$
g) $y = x^{-4}$

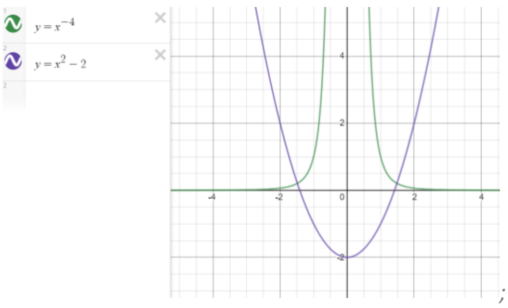




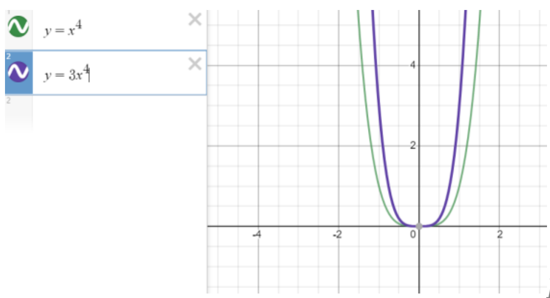




[sudé – b)+ g)



; c)+d)



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4. Vyberte liché funkce a načrtněte jejich graf:

a) $y = x^{-6}$

b) $y = (x+2)^{-3}$

c) $y = x^{4}$

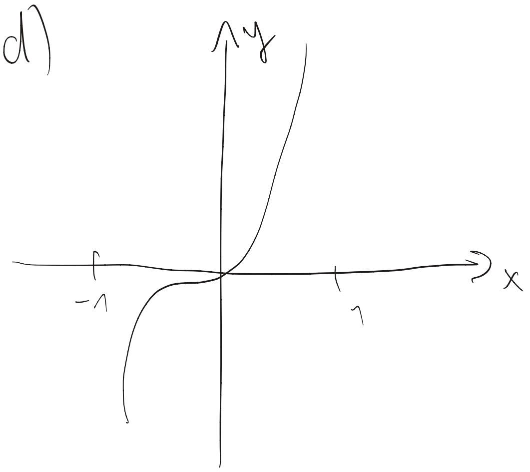
d) $y = 2x^{5}$

e) $y = x^{5}$

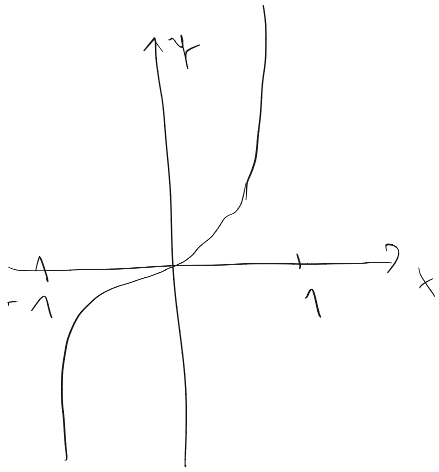
f) $y = x^{-3} - 1$

g) $y = x^{-5}$

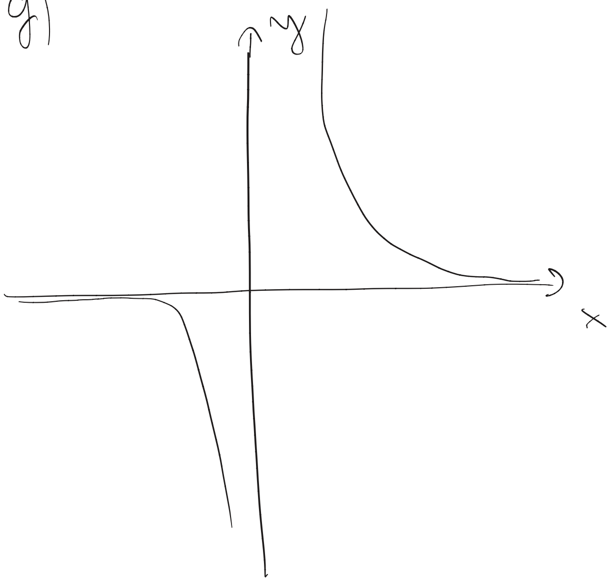
d)



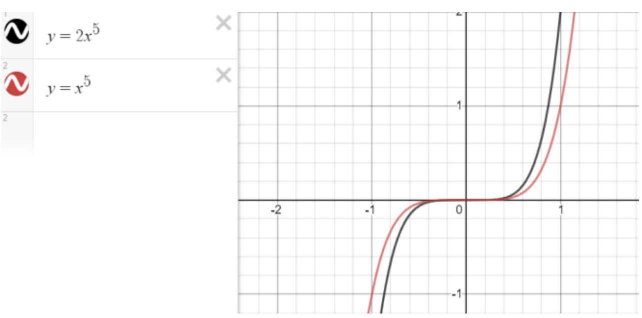
e)



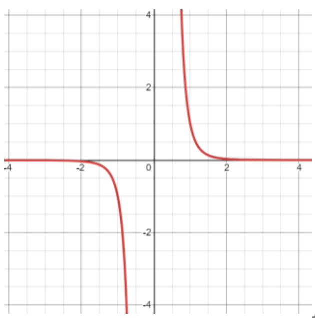
g)



[liché – d) + e)



g)



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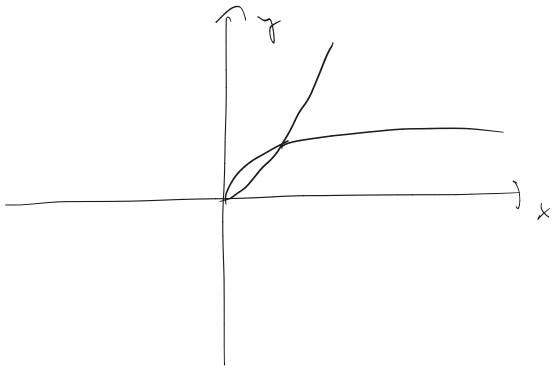
Najděte k zadané funkci inverzní funkci. Načrtněte jejich graf a určete jejich definiční obor.

a) $y = \sqrt{x}$

$D = \langle 0; \infty)$

$f^{-1}: y = x^2$

$D^{-1} = \langle 0; \infty)$

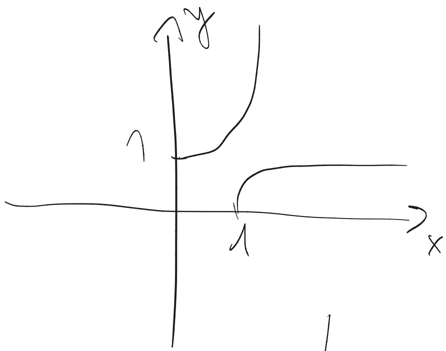


b) $y = \sqrt{x-1}$

$D = \langle 1; \infty)$

$f^{-1}: y = x^2 - 1$

$D^{-1} = \langle 0; \infty)$



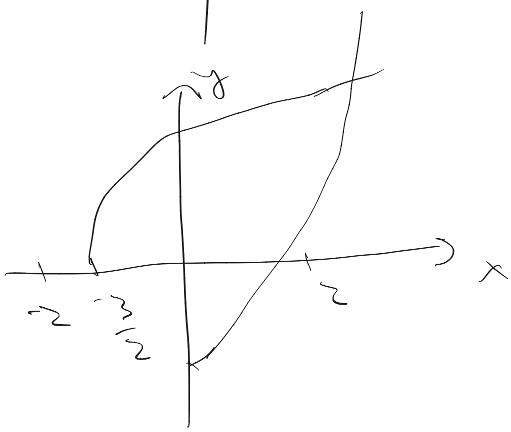
c) $y = \sqrt{2x+3}$

$D = \langle -\frac{3}{2}; \infty)$

$x^2 = 2y + 3$

$f^{-1}: y = \frac{x^2 - 3}{2}$

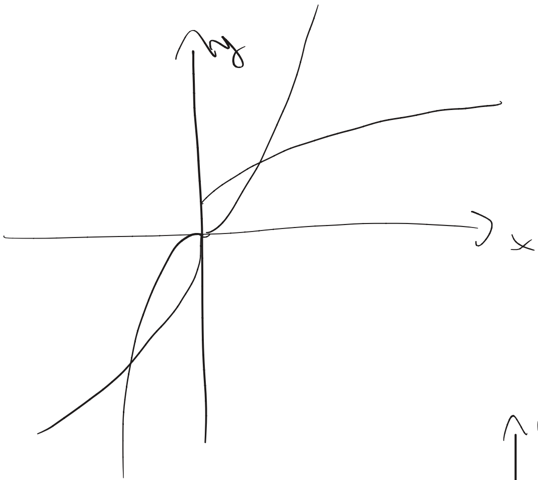
$D^{-1} = \langle 0; \infty)$



d) $y = \sqrt[3]{x}$

$D = R$

$f^{-1}: y = x^3$



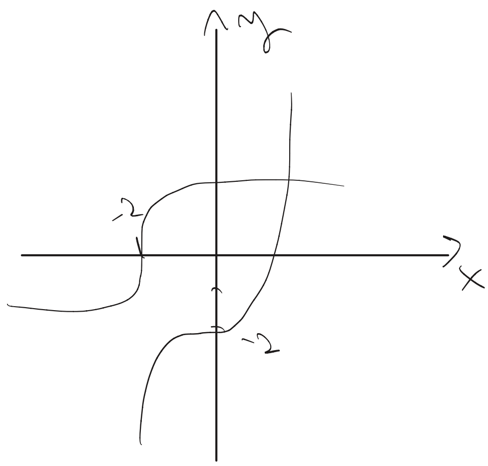
e) $y = \sqrt[3]{x+2}$

$D = R$

$f^{-1}: x^3 = y + 2$

$y = x^3 - 2$

$D = R$



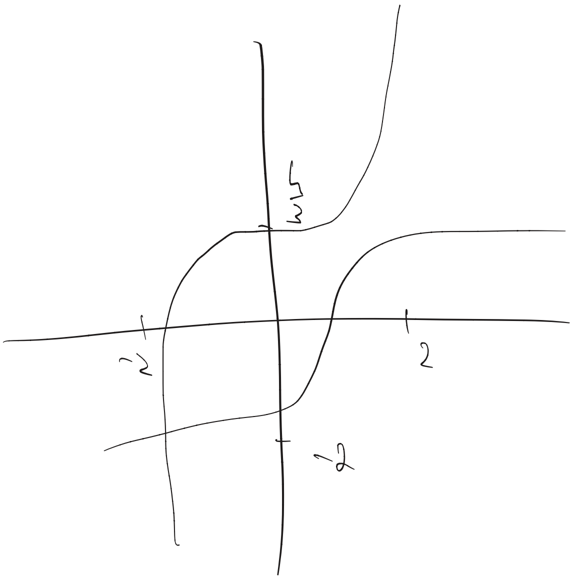
f) $y = \sqrt[3]{3x - 4}$

$D = \mathbb{R}$

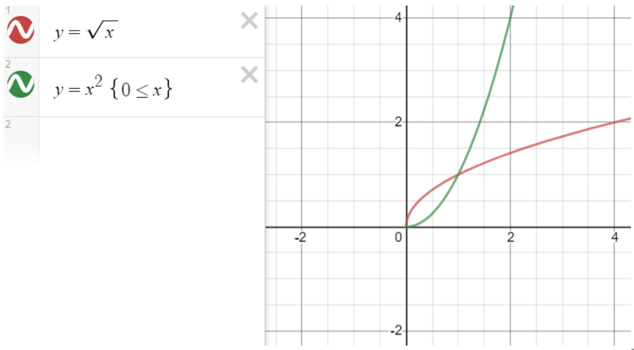
$x^3 = 3y - 4$

$f^{-1}\ \dfrac{x^3+4}{3} = y$

$D^{-1} = \mathbb{R}$



[a) $D(f) = \langle 0;\infty)$; $f^{-1}: y = x^2$; $D(f^{-1}) = \langle 0;\infty)$;



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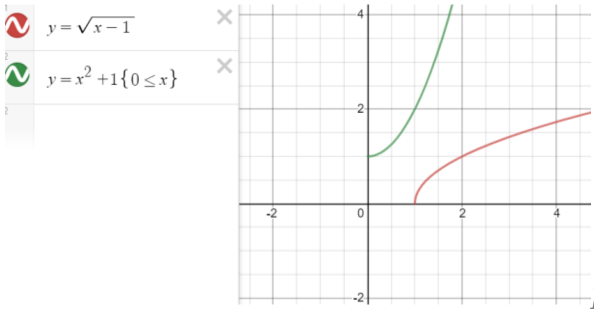
[b) $D(f) = \langle 1;\infty)$; $f^{-1}: y = x^2 + 1$; $D(f^{-1}) = \langle 0;\infty)$;



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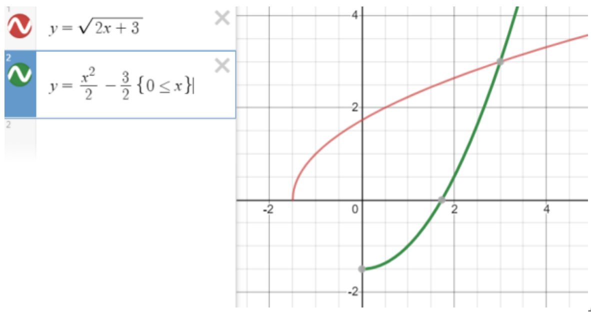
[c) $D(f) = \left\langle -\frac{3}{2};\infty\right)$, $f^{-1}: y = \frac{x^2}{2} - \frac{3}{2}$) $D(f^{-1}) = \langle 0;\infty)$;



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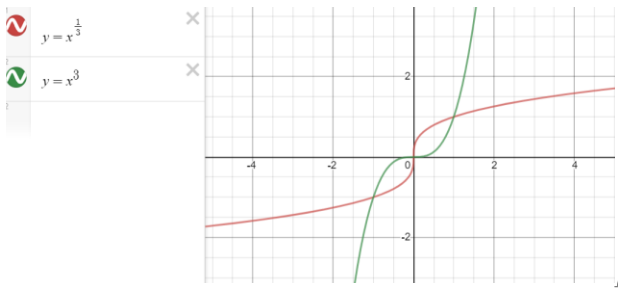
[d) $D(f) = \mathbf{R}$; $f^{-1}: y = x^3$; $D(f^{-1}) = \mathbf{R}$;



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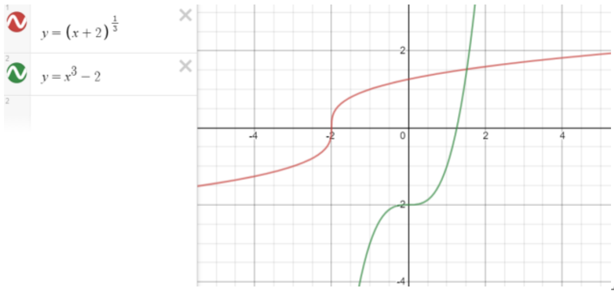
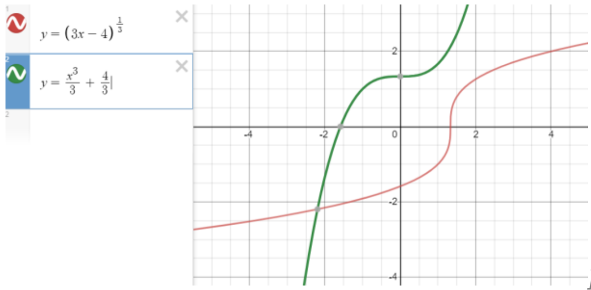
[e) $D(f) = \mathbf{R}$; $f^{-1}: y = x^3 - 2$; $D(f^{-1}) = \mathbf{R}$; ]

[f) $D(f) = \mathbf{R}$; $f^{-1}: y = \frac{x^3}{3} + \frac{4}{3}$; $D(f^{-1}) = \mathbf{R}$; ]

5. Určete definiční obor funkcí:

a) $y = \frac{3}{x^2-1}$

$x^2 - 1 \neq 0$

$x^2 \neq 1$

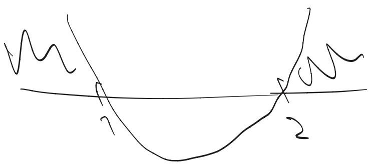
$x \neq \pm 1$

$D = R - \{\pm 1\}$

b) $y = \sqrt{x^2 - 3x + 2}$

$x^2 - 3x + 2 \geq 0$

$D = R - (1;2)$



c) $y = \frac{4-|x|}{3-|x|} - \sqrt{10 - 3x - x^2}$

$3 - |x| \neq 0$

$|x| \neq 3$

(1) $x \neq \pm 3$

$D = \langle -5; 2\rangle - \{-3\}$

(2) $-x^2 - 3x + 10 \geq 0$

$\langle -5; 2\rangle$

*[a) **R** − {−1,1} b) (−∞;1) ∪ ⟨2;∞) c) ⟨−5; 2⟩ − {−3}]*

6. Načrtněte grafy funkcí, určete jejich D(f), H(f), najděte souřadnice průsečíků grafu funkce se souřadnicovými osami, posuďte, zda jsou sudé nebo liché.

a) $y = |x|^{-3}$

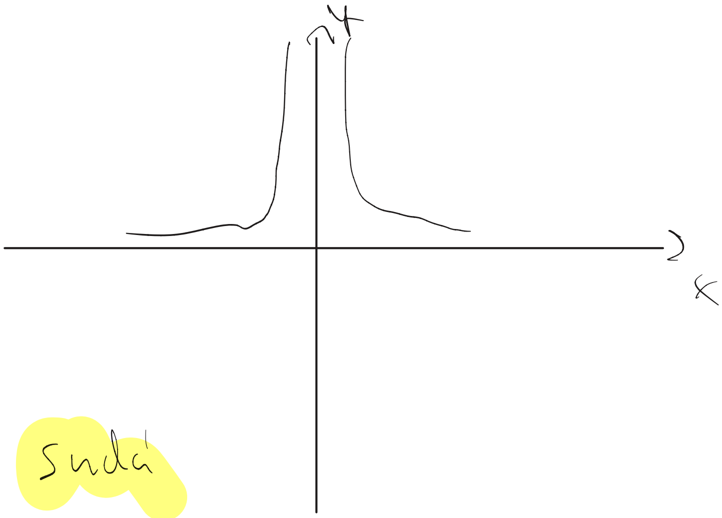
$|x| \neq 0$

$D = R - \{0\}$

$H = (0;\infty)$

P_x není

P_y není



sudá

b) $y = (x-1)^{-3}$

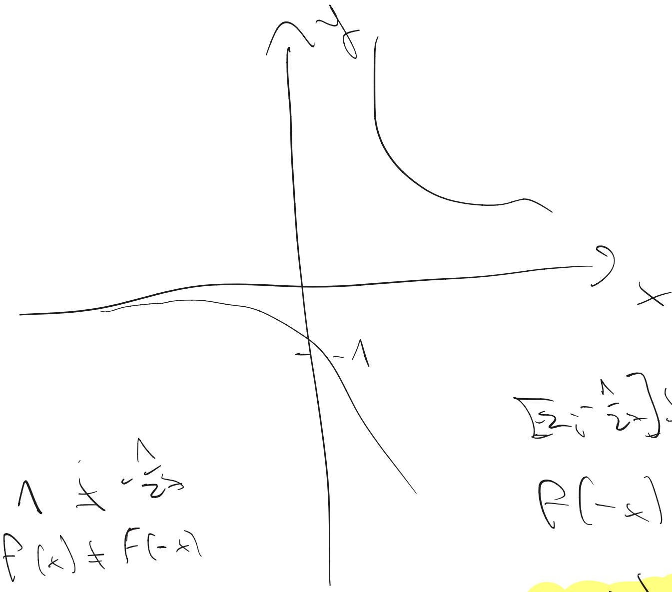
$x \neq 1$

$D = R - \{1\}$

$H = R - \{0\}$

$P_y\ [0;-1]$

P_x není



$1 \neq -\frac{1}{2^3}$

$f(x) \neq f(-x)$

není sudá

$[2; -\frac{1}{2^3}] \neq [-2; 1]$

$f(-x) \neq -f(x)$

ani lichá

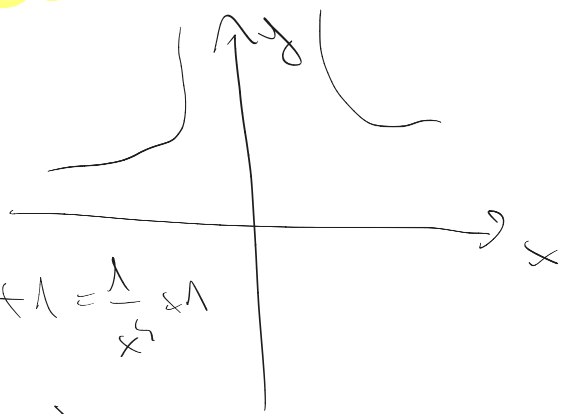
c) $y = |x^{-4}| + 1$

$D = R - \{0\}$

$H = (1;\infty)$

P_y není

$P_x\ [-1;0]$



$\frac{1}{(-x^4)} + 1 = \frac{1}{x^4} + 1$

je sudá

d) $y = -x^{-5} = -\frac{1}{x^5}$

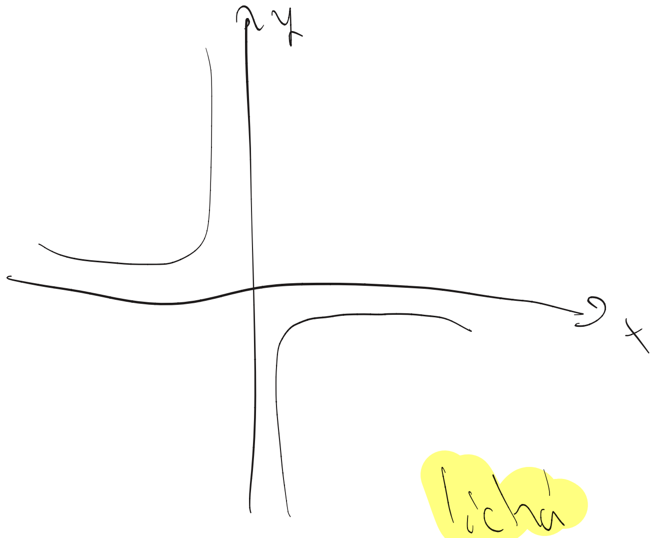
$x \neq 0$

$D = R - \{0\}$

$H = R - \{0\}$

P_x není

P_y není



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e) $y = |x^{-5}| - 1$ $\quad \frac{1}{x^5} - 1$

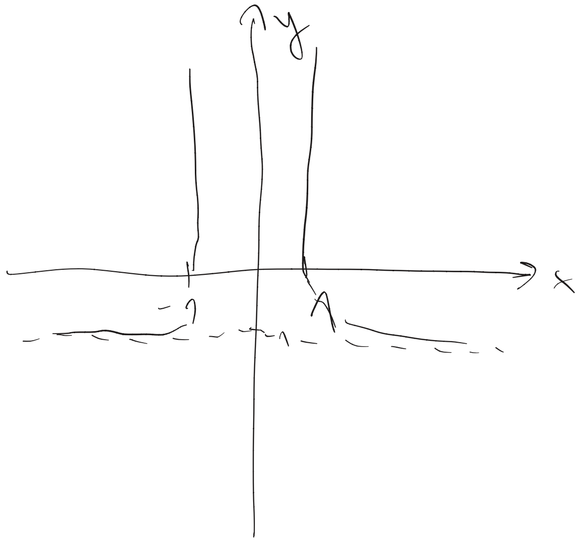
$D = R - \{0\}$

$H = (-1; \infty)$

$P_x\ [0; \pm 1]$

$1 = \frac{1}{|x^5|}$

$x^5 = \pm 1$



P_y není

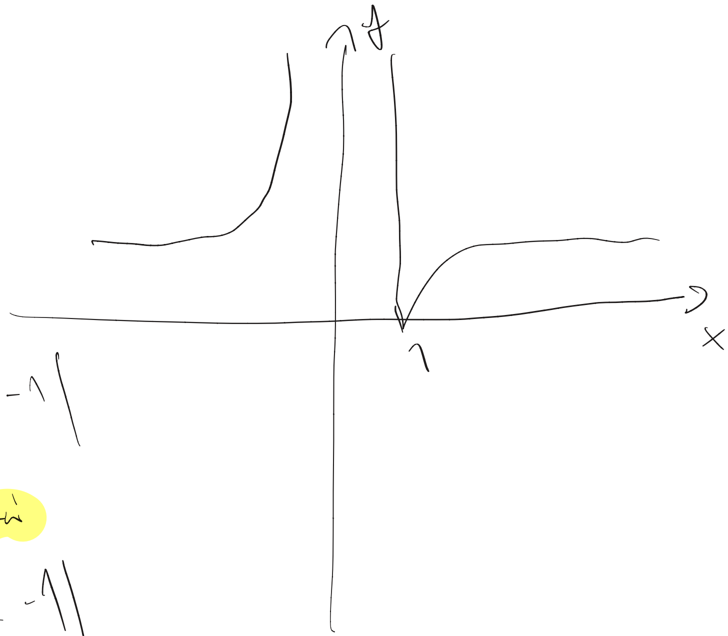
f) $y = |x^{-5} - 1| = \left|\frac{1}{x^5} - 1\right|$

$D = R - \{0\}$

$H = (0; \infty)$

$P_x\ [1; 0]$

P_y není

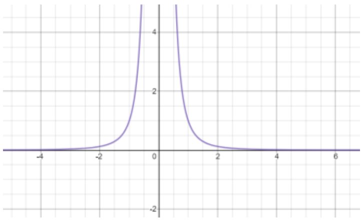


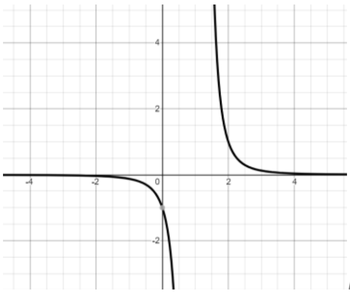
$\left|\frac{1}{x^5} - 1\right| \neq \left|\frac{1}{(-x)^5} - 1\right|$

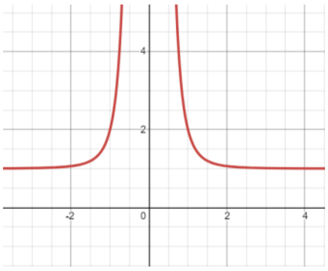
není sudá

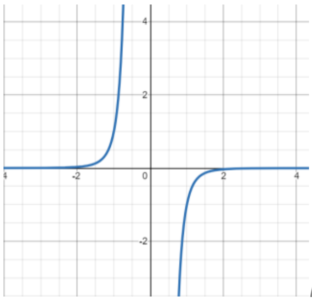
$\left|\frac{1}{(-x)^5} - 1\right| \neq -\left|\frac{1}{x^5} - 1\right|$

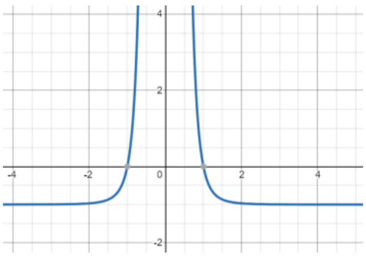
ani lichá

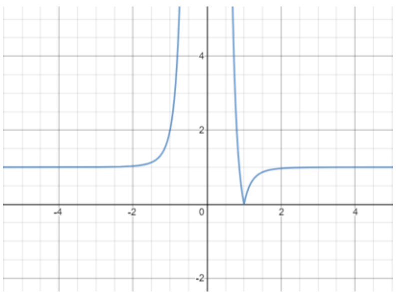
[a)  $D(f) = \boldsymbol{R} - \{0\}$; $H(f) = (0; \infty)$; , P_x, P_y *neex.; sudá fce]*

[b)  $D(f) = \boldsymbol{R} - \{1\}$; $H(f) = \boldsymbol{R} - \{0\}$; P_x *neex.*, $P_y[0; -1]$; *ani sudá ani lichá fce]*

[c)  $D(f) = \boldsymbol{R} - \{0\}$; $H(f) = (1; \infty)$; , P_x, P_y *neex.; sudá fce]*

[d)  $D(f) = \boldsymbol{R} - \{0\}$; $H(f) = \boldsymbol{R} - \{0\}$; P_x , P_y*neex.; lichá fce]*

[e)  $D(f) = \boldsymbol{R} - \{0\}$; $H(f) = (-1; \infty)$; , P_x $[\pm 1{,}0]$, P_y *neex.; sudá fce]*

[f)  $D(f) = \boldsymbol{R} - \{0\}$; $H(f) = \langle 0;\infty)$;, P_x $[1{,}0]$, P_y *neex.* ; *ani sudá ani lichá fce]*

7. Najděte předpis inverzní funkce k funkci f: $y = \frac{2x+3}{x-1}$ funkci. Obě funkce načrtněte. Napište rovnice jejich asymptot.

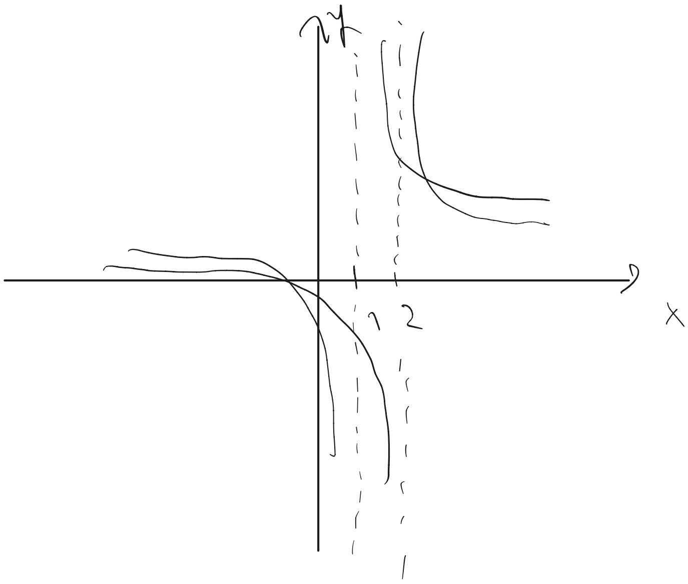
$f^{-1}: x = \frac{2y+3}{y-1}$

$xy - x = 2y + 3$

$xy - 2y = x + 3$

$y(x-2) = x + 3$

$f^{-1}: y = \frac{x+3}{x-2}$



$f(x)$:

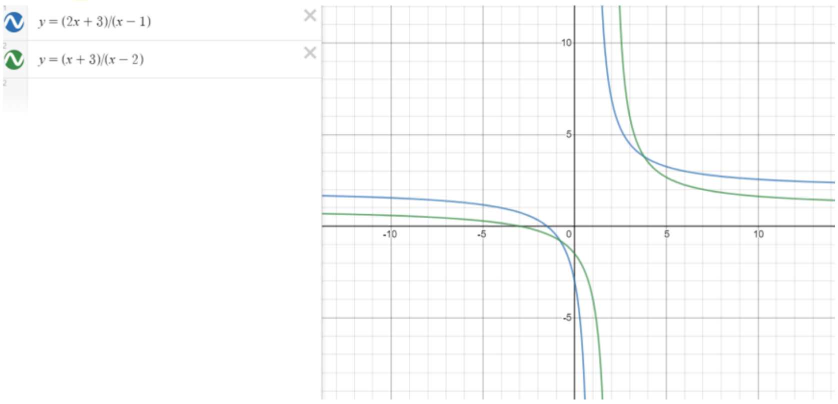
$x \neq 1 \to a_1: x = 1$

$a_2: y = \frac{2}{1} \to 1$

$f^{-1}(x)$:

$x \neq 2 \to a_1: x = 2$

$a_2: y = 1$



$[f^{-1}: y = \frac{x+3}{x-2}$ $as.\ f: x = 1, y = 2;\ as.\ f^{-1}: x = 2, y = 1]$

8. Vyjádřete objem krychle V (m^3) o hraně a (m) jako funkci její tělesové úhlopříčky u (m).

$[V = \frac{u^3 \cdot \sqrt{3}}{9}]$

$V = a^3$

$u = a^2 + (a\sqrt{2})^2$

$u^2 = a^2 + 2a^2 = 3a^2$

$a = \frac{u}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}\,u}{3}$

$a^3 = \left(\frac{\sqrt{3}\,u}{3}\right)^3 = \frac{u^3 3\sqrt{3}}{27} = \frac{u^3\sqrt{3}}{9} = V$